Masking a Logit is not Removing an Option

The case for Gaussian contestant removal over renormalization in machine choice

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Abstract

A softmax renormalizes when a logit is masked, so it obeys Luce's Choice Axiom for that operation. Restricting a choice set in language is a different operation: the prompt changes and the logits are recomputed. We measure which law describes the recomputation. Across thirteen designs and 48,536 cells, scored by exact log probabilities on all but one design, Thurstonian contestant removal predicts restricted machine preference better than renormalization, in all 99 question types of the original appendix and on all twenty-one models. Nothing is tuned, which makes the unit-variance contest a free baseline we recommend in place of renormalization. Much of the reallocation is a flattening that neither unfitted rule anticipates; granting each family one scale parameter leaves the ordering intact. Explicit menus form a separate regime violating every random utility model.

1 Introduction

Large language models can be interrogated cheaply and at scale, which makes the study of their revealed preferences attractive whether one regards them as incipient minds or as statistical mimics of our own predilections. Either way, a question from the psychology of choice arises immediately: when a model distributes probability over alternatives, what law governs how that distribution changes as the set of alternatives changes?

The workhorse answer in economics and psychology is Luce's Choice Axiom [16], which asserts that the probability of selecting item i from a set S is proportional to a fixed utility $u(i)$:

$$P(i \mid S) = \frac{u(i)}{\sum_{j \in S} u(j)}. \tag{1}$$

Its signature is Independence of Irrelevant Alternatives: the ratio $P(A)/P(B)$ is unchanged by the presence or absence of a third option C.

There is a natural alignment between this assumption and the token probabilities of language models, which invariably emerge from a softmax:

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^{n} e^{z_j}}, \tag{2}$$

however complicated the transformer computation producing z_i may be. Delete item 1 by setting $z_1 = -\infty$, a surgical procedure impossible in humans, and the remaining probabilities are the old ones renormalized by $1/(1 - p_1)$. The output layer therefore satisfies the Choice Axiom exactly, for that operation.

Restricting a choice set in language is not that operation, and we do not suppose it is. Telling a model that an option is unavailable changes the prompt, so the whole logit vector is recomputed rather than masked, and nothing in the architecture constrains the new odds among survivors to match the old ones. Menu-dependent logits can represent any positive choice rule at all. The softmax is thus a statement about what the sampling head does with a *fixed* state, not a prediction about behaviour across prompts, and the question of which law describes the recomputation is empirical from the outset.

That gap is the subject of this paper, and it has practical bite because masked renormalization is exactly what deployed systems compute: constrained decoding, tool routers that mask unavailable tools, and guardrails that drop a candidate all impose Equation (1) by construction. What a model reports when told in context that an option is unavailable is a different object. We measure how far apart the two are, and find the difference better described by the Thurstonian contest of [18] than by renormalization.

The evidence is not a single experiment. Table 1 lists thirteen designs comprising 48,536 scored choice cells, built from 61,512 logged exact log-probability measurements on five commercial and twelve open-weight models, alongside the masked language model study of the 2024 version and a generative replication on Claude. Every cell is a zero-parameter out-of-sample prediction: both families are calibrated to an unrestricted distribution and asked to predict a restricted one they have not seen. The two largest batteries, 3,660 and 1,195 cells, favor the contest model with bootstrap confidence intervals excluding zero on every one of the five models, and the margin rises monotonically with model tier, from +0.30 on GPT-4.1-nano to +0.88 on GPT-4.1. All raw model responses are logged and released, so every number below is recomputable without further inference.

Section 2 states the two competing models and Section 2.1 places the comparison in the century of psychometric and econometric argument it belongs to. Sections 3–4 test them on masked language models. Section 5 moves to modern models under exact measurement, using polysemous words as the entry probe and connecting the paradox to the stated-versus-generated gap recently documented by Cekinmez, Wu, Marjieh and Griffiths [19]. Section 6 draws out the implications, and separates what the measurements support from three conjectures they merely motivate: a mechanism for generative "mode collapse", a diversity knob, and a contest-structured output layer.

2 A choice of choice models

One might describe Equation (1) as an *urn model*: item k is represented by balls in an urn in proportion to $u(k)$, a choice is a random draw, and shrinking the choice set merely removes balls. The renormalization of probability under restriction then seems perfectly reasonable.

But perhaps preference is a horse race, not an urn. In the competing model each item i carries a location parameter a_i (a dislikability, say) and an associated performance $X_i \sim N(a_i, 1)$. Item i is chosen when its performance is the lowest draw. This is the model of Thurstone [18], and it makes different predictions about how choice probabilities respond when the choice set is altered: rather than removing a ball from the urn, we remove a horse from the race.

Unit variance and independence make this Case V, the family's default and its tightest special case. Everything below uses it, and Section 5.7 explains why: richer contests would fit

Design	Models	Cells	Verdict
Adjective qualification (masked)	2 masked LMs	100+	Thurstone $\approx$ 80%
Two-slot elimination (masked)	2 masked LMs	100+	Thurstone $\approx$ 80%
Polysemous generation, restriction	Claude	21	re-collapse, not renormalization
Qualified-noun restriction	3 GPT	71	tie (degenerate priors)
Random elicitation, one deletion	3 GPT	77	Thurstone 39/56, $p = 0.002$
Permutation-controlled deletion	5 GPT	1,195	ΔKL +0.025 [+0.018, +0.033]
of which the three headline models	3 GPT	692	ΔKL +0.025 [+0.016, +0.035]
Original 2024 stimuli, two-slot	5 GPT	3,660	ΔKL +0.52 [+0.49, +0.55]
Open-vocabulary sweep, 99 question types	3 GPT	37,764	**99/99 types favor Thurstone**
Ordered selection to depth five	5 GPT	1,363	+0.18 to +0.55, all four framings
Reversibility (best vs worst first)	5 GPT	108	both families fail; not one scale
Block–Marschak / RUM test	5 GPT	30	30/30 violate RUM
Menus, duplicates, decoys	5 GPT	110	substitution best in 15/41; 16/30 regularity
Base against instruct, local	12 open	3,219	tuning effect in 4 of 6 pairs

Table 1: The full evidence base. Cells are scored zero-parameter predictions of a restricted distribution; GPT cells use exact log probabilities rather than sampling. Masked language model counts are category/adjective combinations from the 2024 version.

better, but Luce in its bare form has nothing to fit, so comparing a searched Thurstone against an unsearched Luce would prove little.

(As an aside, it *is* possible to engineer a contest that satisfies Luce's Choice Axiom, by taking $X_i \sim \mathrm{Exp}(h_i)$ with common location and differing hazard rates, or, by Yellott's theorem, with Gumbel noise [27]. The comparison here can therefore be viewed as between two noise families within the contest framework, which is why single-menu fits cannot separate them and the restriction, duplication, and context designs below are required.)

The practical obstacle to Thurstonian models has always been computation, not plausibility. A fast lattice algorithm for calibrating multi-entrant Thurstone models, recovering locations $\{a_i\}$ from winning probabilities and computing derived quantities for fields exceeding 100,000 entrants, was given in [12], and is what makes the experiments below routine.

2.1 A hundred years of the same argument

The question this paper puts to language models is not a new one. It has been contested continuously since Thurstone published the law of comparative judgment in 1927 [18], building on Fechner's discriminal dispersions [1], and it has never been settled in either direction.

Thurstone's account is a contest with normally distributed performances, and its computational awkwardness was recognized immediately: the equal-variance Case V that Mosteller [2] made practical survives because the general case does not reduce to anything convenient. The logistic alternative arrived with Bradley and Terry [3], whose paired comparison model is algebraically tractable exactly where Thurstone's is not. Luce [16] then supplied the axiom, deriving the multinomial form of Equation (1) from Independence of Irrelevant Alternatives and converting a modelling convenience into a behavioral principle.

The objections were immediate and they were about restriction. Debreu [22], reviewing Luce's book, observed that adding a near duplicate of an existing alternative should divide that alternative's share rather than draw proportionally from everything, which IIA cannot do. Tversky [4] showed that similarity between alternatives violates simple scalability altogether and proposed elimination by aspects instead. Luce [6] himself reviewed the axiom's standing after twenty years and treated its empirical failures as established rather than contested. In econometrics the same problem drove a long sequence of repairs: McFadden's conditional logit [5] made the Luce form the workhorse of discrete choice, then nested logit, the generalized extreme value family and mixed logit [10] were built to relax the independence it imposes, with Hausman and McFadden [8] providing a specification test for the violation.

Two results from that literature bear directly on what follows. Yellott [27] proved that Luce's axiom holds in a random utility contest if and only if the noise is double exponential, which places the two families inside one framework and identifies the disagreement as a question about noise rather than about utility. Block and Marschak [25] and Falmagne [26] characterized when a complete choice structure admits any random utility representation at all, by nonnegativity of the Block–Marschak polynomials, which supplies a test that does not presuppose either family. Section 5.11 applies it.

The dispute also has a quantitative history outside psychology. In racing markets, Harville's renormalization [14] is the Luce prediction applied to finishing order, and its systematic mispricing of second and third place is well documented; Henery [7] and Stern [9] proposed normal, that is Thurstonian, order models to correct it, and the correction works. The two-slot design of Section 3 is that comparison transplanted into a language model.

What makes the machine case worth revisiting is that one side of this argument has been built into the apparatus. A softmax output layer is the Luce form, so an LLM's sampling head satisfies the Choice Axiom by construction. Reward models for preference tuning are Bradley–Terry [3], fitted to pairwise human comparisons on the assumption that a single scalar utility governs choice, and Direct Preference Optimization [11] inherits the same likelihood. Listwise ranking objectives use Plackett–Luce [17]. The modern pipeline therefore assumes the Luce side of a century-old and unresolved dispute at three separate points, which makes the question of whether the resulting behavior obeys the axiom an empirical matter with some consequence, and one that exact log probabilities can settle in a way that human subjects never could.

3 Experimental design: masked language models

We probe the token probabilities of BERT [13] and RoBERTa [15], prompted to fill in blanks that invite choices among items of a category. Two designs are used.

In the first, the model answers an *original* question and a *qualified* question whose only difference is an adjective that shrinks the choice set:

1. *Original:* "My favorite state in the U.S. is [MASK] and I try to visit once a year."

2. *Qualified:* "My favorite Western state in the U.S. is [MASK] and I try to visit once a year."

[1]

[1] No qualifier partitions a category cleanly. Some Western states are more western than others, Oklahoma and Texas being the obvious hard cases, and a reader who disagrees with any particular assignment is probably right. The same fuzziness attends every design here: senses of a polysemous word overlap, "a random X" is not a uniform draw, a two-slot continuation invites contrast and collocation as well as exclusion, and an adjective alters what the question means as well as which answers remain admissible. So none of these manipulations is a surgical deletion from a fixed choice problem, and machine restriction is not a pure urn problem in any case. What the designs require is weaker: that the manipulation removes an item from serious contention, and that both candidate laws face the identical stimulus. A qualifier whose boundary is arguable adds noise to the comparison rather than

In the second, we eliminate a single option explicitly, feeding the top answer of a first prompt back into a second:

1. *First:* "My favorite primary color is [MASK] because it is SOMETHING."
2. *Second:* "My two favorite primary colors are [ANSWER] and [MASK] because they are SOMETHING."

where SOMETHING ranges over many adjectives so that luck of phrasing is averaged out (the appendix of the working version lists roughly one hundred category/adjective combinations).

Both models make zero-parameter predictions of the restricted distribution from the unrestricted one. Luce's prediction is renormalization, Equation (1); in the two-step design this is the simplest Plackett–Luce [17] or Harville [14] construction. The Thurstonian prediction calibrates locations $\{a_i\}$ to the unrestricted token probabilities via $\text{Prob}(X_i < X_k \ \forall k \neq i) = p_i$, removes the excluded contestant, and recomputes winning probabilities.

4 Results for masked language models

Table 2 shows the "favorite Western state" prompt for BERT. The Thurstonian column tracks the actual restricted probabilities more closely than Luce renormalization, which overweights the favorite (California) at the expense of the field. Table 3 repeats the exercise for "favorite girl's baby name", where the RMSE using Luce's probabilities is roughly 50% higher.

State	Original (%)	Luce (%)	Thurstonian (%)	Actual (%)
California	10.89	29.87	25.38	20.17
Arizona	6.74	18.50	17.20	10.00
Texas	5.50	15.10	14.59	7.68
Colorado	3.18	8.74	9.38	7.81
Oregon	2.89	7.92	8.67	8.57
Oklahoma	1.97	5.41	6.37	6.64
Nevada	1.66	4.55	5.54	10.18
Montana	1.59	4.37	5.36	14.27
Idaho	1.05	2.87	3.82	4.99
Wyoming	0.98	2.69	3.62	9.69

Table 2: Predicted probabilities under Luce's Choice Axiom and the Thurstonian model for the "favorite Western state" prompt (BertForMaskedLM). Both predictions use only the unqualified token probabilities; the Thurstonian model is closer to the actual qualified probabilities.

Across the full panel of qualified-question and single-item-elimination experiments, the Thurstonian model provides the more accurate prediction in approximately 80% of cases, with its advantage concentrated where the qualification is substantial and the entropy of the unrestricted distribution is low.

favouring either family, and the batteries are large enough, and the restriction phrasings varied enough, that no single fuzzy boundary carries the result. The exclusion phrasing of Section 5.6 is the cleanest instrument, and it is reported separately throughout.

Name	Original (%)	Luce (%)	Thurstonian (%)	Actual (%)
Lily	1.56	20.35	16.18	14.52
Emily	1.15	15.06	13.31	16.29
Mia	0.86	11.25	11.02	13.18
Bella	0.85	11.12	10.93	8.84
Sarah	0.68	8.89	9.44	9.41
Rachel	0.59	7.66	8.56	7.15
Kate	0.59	7.63	8.54	8.44
Lauren	0.47	6.11	7.39	7.23
Brittany	0.46	6.04	7.33	7.55
Chloe	0.45	5.89	7.21	7.39

Table 3: The same comparison for "favorite girl's baby name". The RMSE using Luce's probabilities is approximately 50% higher than the Thurstonian RMSE.

5 Modern models under exact measurement

Cekinmez, Wu, Marjieh and Griffiths [19] recently turned an everyday property of language into a measurable distribution over meanings: give a model a bare polysemous word ("bolt") with no disambiguating context, sample many outputs, and classify which sense each expresses. Across 32 text and image generation models they report a striking ladder of normalized sense entropies: image generation 0.10, text generation 0.25, human subjects 0.47, and, when models are asked to *predict* the distribution of senses, roughly 0.73. Models know the ambiguity; they do not express it. The finding sits within a growing literature on generative homogenization [20] and its sharpening under preference tuning [21]; their own ablation shows DPO alone cutting sense entropy from 0.25 to 0.18.

Read through the lens of Section 2, the ladder is what a contest model produces. Let the stated distribution proxy the latent locations $\{a_i\}$; let each generation be a contest among noisy sense performances. When noise is small relative to the spread of locations, winning probability is a steeply nonlinear function of location differences, sigmoidal rather than convex, and a mildly uneven prior collapses onto its mode. One noise parameter then indexes the entire ladder, and the "multimodal gap" becomes a noise gap rather than a knowledge gap. The rival explanation is a power tilt: renormalized p_i^γ, the Luce family with a temperature. Low-temperature decoding is literally this transformation of a fixed token distribution. Preference tuning and classifier-free guidance are often described in the same terms but are not the same operation: tuning changes parameters, and therefore logits, in a prompt- and token-dependent way, while guidance combines conditional and unconditional logits. The tilt is a rival account of the observed sharpening, not a description of what those procedures compute. The two families are distinguished not by the collapse itself but by behavior under choice-set restriction, precisely the instrument of Section 3, and an experiment run by neither paper until now.

5.1 Replication on a chat model

The smallest of our designs, and the only one using sampling rather than exact probabilities, replicates their generative setup on Claude models (October 2025 generation): 15 polysemous words from the stimulus set of [19], 150 independent sentence generations per word, senses classified by a second model against the closed inventory, plus a stated distribution elicited with their "predicting itself" framing. The collapse replicates almost exactly: mean normalized sense entropy 0.252 generated (their text-model figure: 0.25) against 0.880 stated.

Fitting the two one-parameter families from stated to generated distributions gives a statistical tie (Thurstone $\sigma = 0.42$, RMSE 0.273; power tilt $\gamma = 2.25$, RMSE 0.273; Thurstone the better fit on 10 of 15 words), and the reason both fit poorly is itself informative: for several words the generated mode is not even the stated argmax ("port" generates *harbor* in every sample while rating *connector* most probable). Stated reports are a biased window on the latent locations, not merely a flattened one. This is consistent with [19], whose stated distributions overestimate even human diversity. Calibration should therefore use revealed behavior, which is what the restriction design does.

5.2 Choice-set restriction

For the six words whose unrestricted distributions were non-degenerate, we re-sampled with the modal sense excluded by instruction (150 samples per condition) and compared zero-parameter predictions of the restricted distribution: Luce renormalization against Thurstonian contestant removal, both calibrated to the add-half smoothed unrestricted sense counts.

Word	Luce RMSE	Thurstonian RMSE	Restricted distribution
bolt	0.450	**0.413**	fastener .68, run .18, lock .14
seal	**0.077**	0.089	animal .67, close .33
pitch	0.655	**0.630**	throw .06, tone .31, field .63
bat	0.559	**0.546**	animal 1.00, hit .00
match	0.758	**0.651**	firestick .82, contest .18
iron	**0.002**	0.037	metal .95, press .04, golf .01
pooled	0.495	**0.459**	

Table 4: Restriction test on polysemous words (Claude Haiku 4.5, 150 samples per condition, modal sense excluded by instruction). The Thurstonian prediction is closer pooled and on four of six words (bootstrap over words: $P(\text{Thurstone better}) = 0.90$).

The margin favors Thurstone, but the qualitative signature matters more than the margin: under restriction the model does not spread probability proportionally, as Luce requires. It *re-collapses* onto a new modal sense (match → *firestick* at .82; iron → *metal* at .95; bat → *animal* at 1.00). Remove the winning horse and the runner-up now wins nearly always. That is low-noise contest behavior, in the same low-entropy, heavy-qualification regime where the Thurstonian advantage concentrated in Section 4. Some rank flips under restriction exceed even the Thurstonian sharpening, and inventories with overlapping senses ("bat") contaminate the exclusion instruction; neither family survives those cells.

5.3 Exact probabilities on GPT models

Sampling noise can be removed entirely on models that expose token log probabilities. We ported the qualified-question design of Section 3 to GPT-4o-mini, GPT-4o and GPT-4.1: twenty category pairs ("my favorite color is" / "my favorite warm color is"), two phrasings, next-token distributions read exactly from the top-20 log probabilities, with the Thurstonian field restricted to a declared category inventory. Across 71 usable cells the comparison is a statistical tie with a Thurstonian lean: Thurstone wins 38 of 71 cells and has the lower pooled RMSE on all three models (e.g. 0.290 vs 0.295 on GPT-4o), but a bootstrap over cells does not separate the families ($p = 0.29$).

The instrument, however, has weakened for an interesting reason: the unqualified distributions of preference-tuned chat models are already near-degenerate ("my favorite tree is" puts 0.94 on *oak*), so most cells carry little discriminating information. Under either model the surviving favorite keeps nearly everything, and the comparison hinges on tail probabilities. The masked models of Section 4 answered the unqualified question with entropy to spare (California at 10.9%); their preference-tuned descendants have already collapsed before the experimenter arrives. This is the same collapse measured by [19], now visible as a loss of experimental power: alignment has pushed the interesting variation into the tails, where only exact log-probability designs can reach it.

5.4 Random elicitation restores the regime

The remedy is to ask for randomness rather than preference. "Name a random programming language" followed by explicit single-item elimination raises mean unqualified entropy to 0.38 and returns the experiment to the regime where the two families disagree. The elimination takes one of two forms, either an exclusion ("… that is not Python") or the two-slot continuation of Section 3 ("I already picked one random programming language: Python. Name another…"). The recovery is only partial, since models are poor samplers even on request, with "a random planet" putting essentially all mass on *Mars*. The homogenization of [20] extends to explicit randomness.

To control for phrasing and for which item is removed, the definitive battery deletes *each* of the top eight observed items in turn, under two phrasings, across 50 categories and three GPT models. Thirty-one categories retain enough open-vocabulary mass to be scorable, giving 692 permutation-controlled cells, scored by the proper rule KL(actual $\|$ prediction), with bootstrap confidence intervals over cells.

Stratum	n	mean ΔKL (Luce − Thurstone)	95% CI
all cells	692	+0.025	[+0.016, +0.035]
non-degenerate prior ($\tilde{H} > 0.2$)	521	+0.036	[+0.026, +0.048]
degenerate prior ($\tilde{H} \leq 0.2$)	171	−0.008	[−0.025, +0.011]

Table 5: Permutation-controlled deletion battery, exact log probabilities, KL scoring. Positive favors Thurstone. The Thurstonian advantage is decisive overall and concentrates precisely where the model retains genuine choice entropy.

Contest behavior governs wherever machine preference is genuinely stochastic; the Choice Axiom is competitive only in the degenerate cells that preference tuning has emptied of choice.

5.5 The original stimuli at scale

The largest battery, and the largest effect, comes from re-running the *original* stimulus set of this paper's 2024 version. That is 97 scorable categories crossed with every adjective in the original appendix, 852 distinct prompt pairs, each put to five GPT models in the original two-slot form ("My two favourite human organs are the [ANSWER] and the [MASK] because they are SOMETHING") with exact log probabilities. Across the 2,183 cells of the three headline models the conditional second choice decisively favors the contest model: mean ΔKL +0.55, 95% CI $[+0.51, +0.60]$, and in both entropy strata. Adding the two cheaper models brings the battery to 3,660 cells and +0.52 $[+0.49, +0.55]$. Predicting the runner-up given the winner is precisely where Luce renormalization fails hardest. It is the machine-preference analog of the Harville [14] overpricing of exactas that motivated the Thurstonian computation in the first place. The same questions asked of BERT in 2024 and of GPT-4-class models in 2026 return the same verdict, with a larger margin under exact measurement.

5.6 Every question type, not merely every model

A verdict that holds on average across categories is weaker than one that holds within each of them, so the broadest battery enumerates the design space rather than sampling it. All 99 question types of the 2024 appendix, crossed with all 926 of their adjective variants, under two restriction designs, deleting each of the top eight observed items in turn, on the three cheaper models: 37,764 scored cells from 45,848 logged measurements. Item fields are open-vocabulary, taken from whatever the model itself offers at the unqualified prompt, so no hand-authored inventory can select the categories that suit either family.

Thurstone is ahead in 29,307 of 37,764 cells, a rate of 77.6% that sits almost exactly on the 80% reported for BERT in the 2024 version, now with two orders of magnitude more cells. Mean ΔKL is +0.1495 $[+0.1451, +0.1539]$. Both restriction designs favor the contest model, the two-slot continuation at +0.199 more strongly than the exclusion phrasing at +0.109, and so does every model: +0.111, +0.140 and +0.198 for GPT-4.1-nano, GPT-4o-mini and GPT-4.1-mini.

The breakdown is more informative than the pooled figure. **All 99 question types favor Thurstone** on average, not 90 or 95 of them. There is no category of question, among organs, colors, cities, sports, instruments and the rest, in which renormalization is the better account. The deficit also concentrates where the theory says it must: deleting the model's favorite costs Luce +0.407, against +0.114 for deleting a minor item. Removing the winner is the case Harville [14] mispriced, and it is where the urn and the contest diverge.

At this sample size the degenerate stratum also separates from zero (+0.142 against +0.157 for non-degenerate cells), which sharpens the earlier reading. Preference tuning does not repeal the contest structure in collapsed distributions; it only leaves too little entropy for a smaller battery to detect it.

Cells nest inside question types, adjective variants, base prompts and models, so cell-level bootstrap intervals overstate the independent information. A cluster bootstrap that resamples the 99 question types and retains every cell belonging to a selected type gives $[+0.135, +0.165]$ against the cell-level $[+0.145, +0.154]$. The clustered interval is the honest one and is reported throughout; here it is barely wider, because the effect is present within essentially every cluster rather than concentrated in a few.

5.7 What the comparison does and does not identify

One convention governs every comparison in this paper and should be stated before the numbers are read. The contest used throughout is independent performances of equal unit variance, which is Thurstone's Case V, the default member of the family and the one Mosteller [2] made practical.

It is also the most constrained member. The family contains freer relatives, among them unequal variances, correlated performances and heavy-tailed noise, and any of them would fit better by construction.

Searching that space and reporting the winner would be unfair to Luce, which in its bare form has nothing to fit. So nothing is tuned. Every prediction in every battery is computed from the unrestricted distribution alone, with no free parameter and no training set, which is what makes the cells out-of-sample in the strong sense: neither family ever sees a restricted distribution before predicting it. The price of that discipline is that a positive result cannot be attributed to Gaussian geometry specifically. It shows that the default contest forecasts better than the default urn.

Nor does winning that comparison establish that a Gaussian race generates the data. The obvious alternative account is one-parameter sharpening,

$$\tilde{p}_i(\gamma) = \frac{p_i^{\gamma}}{\sum_{j\in S} p_j^{\gamma}}, \qquad i \in S, \tag{3}$$

which is the Luce family with a temperature. Fitting a single global γ on some question types and scoring it on entirely held-out types, so that no category is ever used for both, gives $\gamma \approx 0.22$ and a held-out mean KL of 0.803, against 1.786 for zero-parameter contestant removal and 1.926 for plain renormalization.

The tilt is not a rival forecast and is not reported as one. It cannot be computed from an unrestricted distribution alone: fixing γ requires a corpus of already-observed restrictions to train on, which is precisely the information the zero-parameter designs withhold. Read instead as a diagnostic, it says something useful about the residual. The fitted γ is well below one, so restricted distributions are markedly *flatter* than the unrestricted ranking implies, and both zero-parameter forecasts are too concentrated. A large part of the reallocation is a change of scale, and no zero-parameter rule can know that scale in advance.

The question the diagnostic then raises is whether Gaussian geometry survives once the scale is granted, and the answer requires granting it to both sides. Performances $N(a_i, \sigma^2)$ with a single σ fitted on training question types and scored on held-out ones give $\sigma \approx 3.4$ and a held-out mean KL of 0.794, against the tilt's 0.803: an advantage of $+0.0089$ $[+0.0040, +0.0142]$ clustered by question type, favouring Thurstone in 60 of 99 types. The ordering is unchanged at one parameter each, and both are far better than either family unfitted.

The defensible reading follows. Prompted restriction reallocates probability rather than renormalizing it. Among forecasts that tune nothing, contestant removal beats renormalization everywhere we look; a large part of what neither captures is a flattening whose magnitude has to be learned from data; and when each family is allowed to learn it, the contest still comes out slightly ahead.

The practical recommendation is therefore about benchmarking. Case V contestant removal costs nothing to compute, requires no training data and no fitted quantity, and predicts restricted machine choice better than renormalization in every design here, on every model, and in all 99 question types. It is the stronger default, and work on machine choice modelling should be measured against it rather than against renormalization. A method that merely improves on Luce has cleared a low bar. Whether the same holds for human choice is a separate question that this paper's data cannot address.

What the batteries do not establish is that a stable Gaussian contest runs inside the model. Several untested alternatives would produce similar reallocation, among them heteroscedastic or correlated races, heavy-tailed contests, similarity-sensitive nesting, and retrieval dynamics specific to list completion.

5.8 The deficit grows with model tier

Across the five models the margin is ordered by tier: GPT-4.1-nano +0.30, GPT-4o-mini +0.35, GPT-4o +0.42, GPT-4.1-mini +0.65, GPT-4.1 +0.88. The more capable the model, the further its revealed preferences sit from the Choice Axiom its own output layer suggests.

The architecture is constant along that ladder, which narrows the candidate explanations without settling them. All five models emit a softmax, and a softmax implements Equation (1) mechanically: mask a logit, renormalize, and the Choice Axiom holds by construction. The deviation therefore cannot originate in the output layer. It originates in what the network computes before the output layer, and it grows with capability, as though the hidden layers were working against their own parameterization and getting better at it.

The obvious confound is concentration. More capable models answer the unqualified question more sharply (mean $\tilde{H}$ falls from 0.276 on GPT-4.1-nano to 0.105 on GPT-4.1 over matched prompts), and sharpness alone widens the gap between an urn and a contest, since the contest map amplifies rank separation. Three controls bear on the two explanations, all run on the 512 prompts measured on every model, paired within the identical question. None of them holds concentration fixed continuously, so the stratification below is a coarse control rather than a matched comparison.

Paired comparison (same question prompt)	n	ΔKL difference
GPT-4.1 − GPT-4.1-nano	512	+0.571 [+0.426, +0.721]
GPT-4.1 − GPT-4o	512	+0.478 [+0.348, +0.613]
GPT-4.1-mini − GPT-4o-mini (same size class)	512	+0.276 [+0.172, +0.380]
GPT-4.1 − GPT-4.1-nano, both degenerate	142	+0.717 [+0.387, +1.067]
GPT-4.1 − GPT-4.1-nano, both non-degenerate	144	+0.363 [+0.191, +0.548]

Table 6: The tier effect survives matching on the question and on concentration. Positive means the more capable model departs further from Luce renormalization.

Holding the question fixed, holding concentration fixed, and holding the size class fixed while advancing one model generation all leave the effect intact. Sharper locations are therefore part of the story but not the whole of it: at matched concentration the frontier model still departs further from renormalization than the cheapest one.

The panel admits one further decomposition, into a generation axis and a scale axis, and the two do not behave alike. Advancing a generation at fixed size class moves the deficit reliably, by +0.478 [+0.348, +0.613] for the flagships and +0.276 [+0.172, +0.380] for the mini class. Varying scale within a generation moves it too, by +0.253 and +0.318 for the two steps down the 4.1 family, but the corresponding step within the older family, GPT-4o against GPT-4o-mini, is +0.050 [−0.037, +0.140] and does not exclude zero. Within that generation, in other words, making the model larger left its departure from the Choice Axiom essentially unchanged, while moving the same size class one generation forward changed it substantially. Whatever separates the two generations appears both to raise the deficit and to make it sensitive to scale.

We report this as a pattern in five models rather than a scaling law, and note two limits. Size class here is a vendor label, not a parameter count, and tier is a proxy for capability rather than a measurement of it. The pattern does, however, yield a falsifiable prediction: if what separates the generations is post-training rather than scale, a base model should obey the Choice Axiom more closely than its instruction-tuned sibling. Section 5.9 tests that prediction directly, on models whose base and instruction-tuned checkpoints are both available.

5.9 Preference tuning moves the deficit, in some families

The commercial models expose no base checkpoints, so this test uses open-weight pairs run locally, where the whole vocabulary is available and nothing is truncated. Base and instruction-tuned siblings receive the identical completion prompt, with no chat template on either side, so the checkpoint is the only difference. Both are given exactly the stimuli and the design of the sweep in Section 5.6, so that nothing about the measurement varies either.

Contrast (paired within question, adjective and deleted item)	n	ΔKL step	95% CI
Instruction tuning, Qwen2.5 0.5B	751	+0.0026	[+0.0019, +0.0034]
Instruction tuning, Qwen2.5 1.5B	627	+0.0038	[+0.0017, +0.0060]
Instruction tuning, Qwen2.5 7B	809	+0.0247	[+0.0193, +0.0305]
Instruction tuning, Mistral 7B	171	+0.0194	[+0.0122, +0.0286]
Instruction tuning, gemma-2 2B	453	−0.0003	[−0.0008, +0.0001]
Instruction tuning, Llama-3.1 8B	408	−0.0010	[−0.0016, −0.0004]
All six pairs pooled	3,219	+0.0084	[+0.0069, +0.0100]
Scale, base only, 0.5B to 1.5B	410	+0.0025	[+0.0018, +0.0033]
Scale, base only, 1.5B to 7B	484	+0.0017	[−0.0000, +0.0033]
Scale, base only, 0.5B to 7B	261	+0.0039	[+0.0022, +0.0055]

Table 7: Instruction tuning against the matching base checkpoint, and scale among base checkpoints. Positive favors Thurstone. The tuning effect is large in two families, absent in a third, and slightly reversed in a fourth.

The prediction holds in four of the six pairs and fails in two. Tuning moves Qwen2.5 away from Luce at all three sizes, by an amount that grows with scale from +0.0026 to +0.0247, and moves Mistral 7B by +0.0194. It does nothing detectable to gemma-2 2B, and it moves Llama-3.1 8B slightly the other way, by −0.0010 with an interval excluding zero. Pooling all six gives +0.0084 [+0.0069, +0.0100], but that pooled figure rides on Qwen and Mistral and should not be read as a general law.

Where tuning does move a model, it moves it further than scale does. Among Qwen2.5 base checkpoints the entire span from 0.5B to 7B is +0.0039, about a sixth of what tuning contributes at 7B alone. So post-training is capable of dominating parameter count in setting how far a model sits from the Choice Axiom, which is the mechanism the commercial generation effect of Section 5.8 suggested. But the two null and reversed families show that it does not do so by necessity, and whatever distinguishes Qwen and Mistral post-training from Llama and gemma post-training is not visible from outside the labs.

Preference tuning is therefore one route by which a model departs from renormalization, substantially so in some families, and the direction of the effect is not fixed across families. That is consistent with the DPO ablation of [19], which changes sense entropy with no change in size, without implying that every tuning recipe pushes the same way.

Three further caveats bound all of these numbers. The absolute magnitudes are one to two orders of magnitude below the commercial panel on identical stimuli, so the local and commercial ladders are separate evidence rather than one curve. The weights are 4-bit quantized, which could compress the tails where the families disagree. And two of the six pairs sit so close to zero deficit overall that the paired contrast has little room to move in either direction. Whether Thurstonian diagnostics can serve as a label-free measure of model quality is a natural next question, and Section 5.10 counsels caution, since the reversibility diagnostic does not order the models by capability at all.

The practical import falls on anyone who removes an option and renormalizes. Constrained decoding, tool routers that mask unavailable tools, and guardrails that drop a candidate all compute the Luce prediction, because that is what masking a logit does. What the model reports when told in context that an option is unavailable is measurably different, and the discrepancy is largest on the most capable models.

One further caution emerged with practical import for all such experiments: the "actual" restricted distribution depends materially on how the restriction is phrased. Holding everything else fixed, the exclusion and two-slot variants disagree about the restricted distribution by a median maximum gap of 0.14, and up to 0.65 (GPT-4o-mini names *Earth* after *Mars* with probability 0.94 under one phrasing and 0.56 under the other). Under surgical removal from a fixed preference state the two phrasings would coincide; in a language model the choice-set manipulation is itself a prompt intervention. Framing effects of this size sit above both choice models, and any account of machine preference that ignores them is incomplete.

5.10 Ordered selection, depth, and reversibility

The two-slot design is an exacta. Asking instead for an ordered list, and eliciting each slot separately with the earlier picks fed back in, extends it to a trifecta and beyond, so that Plackett–Luce [17] renormalization and Thurstonian contestant removal can be compared at every depth. The contest calculation here is stagewise: the taken items are removed and the surviving field is re-raced, which is not the conditional order statistic of a single Thurstone draw. It suits successive prompts, where the model recomputes its distribution at every slot, but the racing vocabulary should not be read as the one-draw joint ranking model. Four framings were run over 40 categories and five models to depth five: two neutral ("select five random X in order", "name five different X, in order") and two preference-ordered (most favourite first, least favourite first).

The contest model wins under all four framings, which are different elicitation architectures rather than paraphrases: +0.55 most-favourite-first, +0.38 different-in-order, +0.26 random-order, +0.18 least-favourite-first, all intervals excluding zero over 1,363 cells.

Depth, however, does not behave as the racing analogy predicts. Since slot d involves $d-1$ removals and renormalization can only redistribute vacated mass proportionally, the Luce deficit might be expected to compound with depth. It does the opposite, falling from +0.42 at slot two to +0.24 at slot five, and within a given category and model the change from the second slot to the deepest is negative on average (−0.23, positive in only 161 of 395 ladders). The mechanism is visible in the cell sizes: each removal shortens the surviving field, and by slot five most cells have only two or three candidates left, where the two families cannot differ much. Depth in an open-vocabulary elicitation removes discriminating power rather than adding it, which is the opposite of the racing case, where the field stays large as places are filled.

Reversibility is a sharper test, and both families fail it. Under a Thurstonian scale, asking for the least favourite is the same latent vector read backwards, since the minimum of X is the maximum of $-X$; one calibration must then predict both directions with no free parameters. Luce's axiom, by contrast, says nothing about worst choice; replacing weights w_i by $1/w_i$ is an extra bipolar assumption rather than a consequence of IIA, and negating a Gumbel shock does not give another Gumbel shock. The two sides of this test are therefore not symmetric. Neither works: predicting the least-favourite-first distribution from the most-favourite-first one costs 5.3 to 9.0 nats depending on the model, for both families alike, and the ranking of models by this quantity does not track capability. The reason is that the two directions barely share an item set. Asked for a least favourite, models name items they never mention when asked for a favourite, so the reversed question is not the same contest re-read but a different one. Machine preference in this regime is not a single scale with two ends.

5.11 Menus, duplicates, and context effects

Presenting the choice set explicitly changes the phenomenon entirely, in three ways. The design is "pick one of these: …", with menu order randomized and probabilities read from option-token log probabilities.

First, menu choice by preference-tuned models is nearly deterministic. A transport menu (car, bus, train, bike, walk, taxi) under contextual conditioning behaves with perfect semantic sense. "It is raining" sends GPT-4o to *taxi* with probability 1.00 and "there is a fuel shortage" sends it to *bike* at 0.99. But the resulting distributions are so degenerate that menu-based deletion cannot separate the choice models (144 cells, ΔKL CI $[-0.028, +0.016]$), consistent with Table 5's stratification. The determinism extends to simulated agents: fictional election candidates with platform descriptions, evaluated through fictional voter personas, produce near-certain votes even when the persona is written to be genuinely cross-pressured, and candidate-withdrawal cells are consequently uninformative. The model represents a persona as an argmax decider; it does not carry within-persona randomness at all.

Second, machines are anti-red-bus. In Debreu's classic objection to the Choice Axiom [22], splitting a bus into a red bus and a blue bus should not double the bus vote; humans treat near-duplicates as substitutes, which correlated-noise (probit) models capture. Across ten duplicate-pair families the models show little sign of substitution: the duplicated category's total share typically inflates, often beyond even the urn-doubling prediction (apple $0.08 \to 0.53$ when split into green and red), and a substitution baseline is the best predictor in only 15 of 41 cells against Luce's 24. (With exchangeable duplicates, Luce and independent Thurstone nearly coincide; the test's power is entirely against substitution.) Description salience, in that a *red* apple is more vivid than an apple, appears to dominate any shared-noise structure.

Third, where a menu does leave residual entropy, machine choice violates *regularity*, the property, shared by every random utility model including Thurstone's, that an added option cannot increase an incumbent's absolute share. Using two-attribute options (price versus quality) in randomized lettered menus, an asymmetrically dominated decoy [23] increased the target's absolute share in 16 of 30 family-model cells, and a compromise manipulation [24] in 7 of 30. The phone-plan family shows a textbook attraction effect (GPT-4o: target share $0.26 \to 0.84$ when its decoy appears). Machine preference, like human preference, is not a random utility model in the menu regime.

The menu regime admits a verdict stronger than any pairwise model comparison, because exact probabilities make it feasible to measure the *complete choice structure*: all eleven menus over a four-item universe, a measurement no human experiment delivers cleanly. By the Block–Marschak–Falmagne theorem [25, 26], such a structure is random-utility representable if and only if its Block–Marschak sums are nonnegative. Across six item families and five GPT models, **all 30 measured structures violate the conditions** (4–9 negative terms of 32 each; worst

−0.77, far beyond measurement noise from order averaging). Fitting the structures anyway, the best possible random utility model (a distribution over all 24 rankings) leaves half the divergence unexplained (total KL 8.1, against 16.2 for the best Thurstone locations and 17.5 for the best Luce utilities, with Thurstone the better parametric fit in 26 of 30 structures). In the menu regime, machine choice is not a random utility model at all.

Together the batteries locate the choice law in the *elicitation architecture*. Open-vocabulary recall behaves like a Thurstonian contest among retrieved candidates (Table 5); explicit menus behave like near-deterministic maximization perturbed by salience and context effects that no random utility model accommodates. Between them sits the sampled-sentence regime of [19], where the contest reading again does best.

6 Discussion

These results concern machines, and contain no human choice data; similarity of outputs would not identify similarity of mechanism, so nothing here bears on whether people obey the Choice Axiom. What they do suggest is that the axiom's standing as a default for machine choice owes more to computational convenience than to fit, a convenience the algorithm of [12] removes.

The polysemy results admit a further conjecture, offered as such. Generative "mode collapse" may be less a defect of knowledge than the output of a low-noise contest run over intact locations, in which case diversity would be restored not by a power tilt but by inverting the contest to recover locations and re-sampling at a chosen noise level, using the calibration of [12] and no retraining. The evidence for this is a fifteen-word exploratory design with six usable restriction cells, and it shows that a contest transformation approximates the sense ladder, not that generation is literally a race over a fixed inventory of senses. Establishing the remedy would require a held-out intervention study setting Thurstone-noise adjustment against temperature, top-p, typical and diversity-aware decoding, scored on both diversity and quality. We have not run it.

The same applies, more strongly, to architecture. That contextual restriction follows contestant removal more closely than renormalization suggests a contest-structured output layer might generalize better, and no experiment here tests generalization at all. A contest link would change subset behaviour within a fixed state; it would not by itself make upstream scores invariant across prompts, which is where the framing effects of Section 5.6 live. What follows is a feasibility note rather than a result.

Current architectures oblige the hidden layers to work against the renormalization their output layer implies, and Section 5.8 indicates they do so more, not less, as models improve.

The usual objection is cost, and it is weaker than it looks. Winning probabilities in the normal contest have no closed form, but they reduce to a single shared one-dimensional integral,

$$P(X_i \text{ minimal}) = \int \phi(x - a_i) \prod_{j \neq i} \bigl(1 - \Phi(x - a_j)\bigr)\, dx, \tag{4}$$

and the leave-one-out product it contains is the quantity the lattice method of [12] is built to avoid recomputing. Discretize the ability axis, form the survival function of each contestant on the common grid, and accumulate prefix and suffix products across the field. Every contestant's leave-one-out product is then a single elementwise multiplication of the prefix ending before it and the suffix beginning after it, so the entire vector of winning probabilities costs two passes over the grid rather than n separate products, which is $O(nG)$ for n items and G grid cells instead of the $O(n^2G)$ the expression suggests. At the default resolution of that method G is a few hundred cells.

The consequence that matters for training is that the same prefix and suffix arrays already contain the derivatives. Differentiating Equation (4) with respect to any a_j reuses the products computed for the forward pass, so the backward pass costs the same order as the forward one and there is no autodiff graph over the $n(n-1)$ survival terms to build or traverse. Removing one contestant, which is the operation the whole paper measures, is also the operation that short-circuits the gradient.

So the layer is a training-time proposition rather than an inference-time curiosity. The contest can enter the update rule itself, with the network learning locations $\{a_i\}$ against a contest likelihood instead of logits against a renormalization likelihood, which is the arrangement Section 5.8 suggests the hidden layers are already straining toward. The remaining constant, a few hundred grid cells against a softmax's single pass, is a fused kernel of two elementwise sweeps and one convolution, and it can be cut further by letting a softmax shortlist the top hundred candidates and the contest arbitrate among those, which is where essentially all of the mass and all of the measured disagreement between the two families lives.

We have not built it. Equation (4) and the leave-one-out structure are offered to show that what stands in the way is a constant factor and an implementation, not a missing gradient.

The limitations are specific rather than general. The generative replication of Section 5 is the smallest design here, and the only sampled one: fifteen words, one model, a single machine judge, sampling temperature uncontrolled. The full stimulus set and model panel of [19] would settle that one-parameter comparison with real power. The local panel is 4-bit quantized, and its magnitudes cannot be pooled with the commercial panel. Reversibility remains unexplained by either family. None of these touch the central measurement, which is that on 48,536 scored cells, thirteen designs, twenty-one models and all 99 question types of the original appendix, the contest predicts restricted machine preference better than renormalization does.

Data and code availability

Every battery, its stimuli, and every model response are in `research/polysemy_pilot` of `github.com/microprediction/winning`, which also contains the Thurstone lattice implementation used throughout. Each paid API response is written to an append-only JSONL log before anything is computed from it, so all derived numbers are recomputable offline: the logs total 61,512 measurements (`sweep_raw.jsonl`, `vol_raw.jsonl`, `perm_raw.jsonl`, `random_raw.jsonl`, `exact_raw.jsonl`, `bm_raw.jsonl`, `local_qvols_raw.jsonl` and others), and re-running any battery script re-scores from the logs without further inference. The item inventories are in `inventory.py`, the model tiers in `models.py`, and the tilt and clustered-bootstrap analyses in `power_tilt.py` and `one_param.py`.

Settings common to the commercial batteries: chat completions with `max_tokens=1`, `logprobs=True`, `top_logprobs=20`, `temperature=1.0`, no penalties, no seed. Models are the aliases `gpt-4o`, `gpt-4o-mini`, `gpt-4.1`, `gpt-4.1-mini` and `gpt-4.1-nano`, queried in August 2026; the logs record each response verbatim, but the aliases are moving targets and a future re-run need not reproduce them exactly. The GPT-5 tier cannot be used at all, since it refuses log-probability requests. Local models are 4-bit `mlx-community` conversions run with full-vocabulary logits.

Candidate handling, which matters for the comparison: item fields are the top-ten surviving tokens after lower-casing, stripping, and discarding non-alphabetic and stop-word tokens;

probabilities of surface variants that normalize to the same word are summed; the retained mass is renormalized over the field, so an item's probability is its first-token mass rather than a full-sequence probability. Truncation is not neutral between the two families. Under renormalization, conditioning the winner to lie in a retained set equals deleting the omitted alternatives, whereas under a contest the two differ, so a Thurstone calibration fitted to renormalized top-k probabilities need not recover the locations of a coherent full-vocabulary contest. The local batteries avoid this by scoring whole items by teacher forcing over the entire vocabulary. Cells are discarded when the field holds fewer than three items, when fewer than two survive the restriction, or when the calibration residual exceeds 0.05; mass appearing at the restricted prompt on items absent from the unrestricted field is outside both models' support and is excluded from the renormalization rather than smoothed.

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